Harmonics

If a signal is periodic with frequency *f*, the only frequencies composing the signal are integer multiples of *f*, i.e., *f*, 2*f*, 3*f*, 4*f*, etc. These frequencies are called **harmonics.** The **first harmonic** is *f*, the **second harmonic** is 2*f*, the **third harmonic** is 3*f*, and so forth. The first harmonic (i.e., *f*) is also given a special name, the **fundamental frequency**. Figure 11-7 shows an

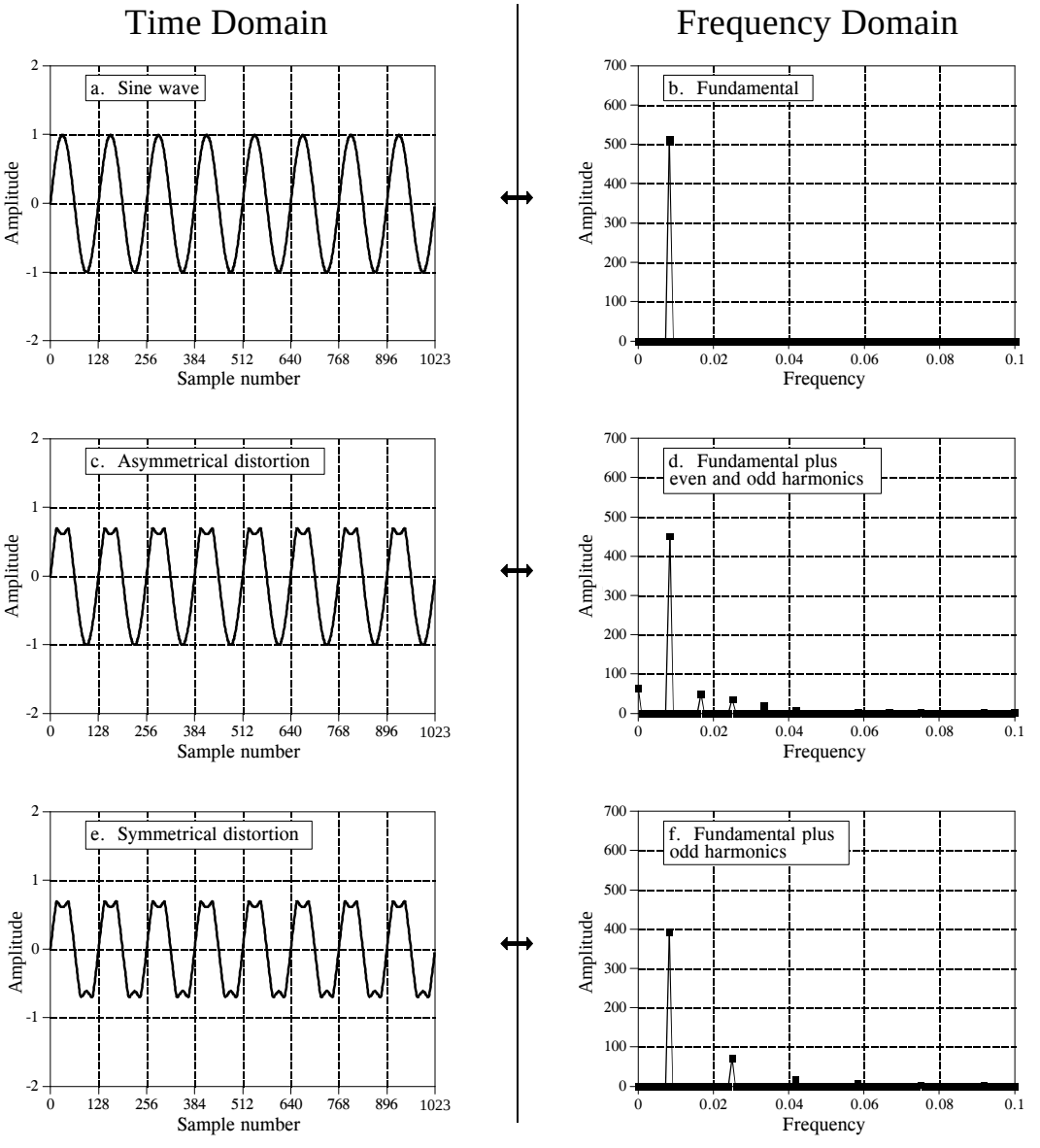


FIGURE 11-7
Example of harmonics. Asymmetrical distortion, shown in (c), results in even and odd harmonics, (d), while symmetrical distortion, shown in (e), produces only even harmonics, (f).